

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Pharm Examination

University Theory Examination April 2016

PH304 Pharmaceutical Biochemistry- I

Date: 11.04.16 , Monday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q- 1 Attempt any TWO of the following:

- A I Define: (i) Coenzymes (ii) β -oxidation (iii) Ketonemia (iv) Saponification
Number (v) Essential fatty acids (vi) Rancidity 6
- II Classify different biomolecules with suitable examples. 4
- B I Write down short note on biosynthesis of long chain fatty acid. 6
- II Write about the significance of enzymes in diagnosis of disease. 4
- C I Write a note on enzyme inhibition. 6
- II Define and differentiate Eukaryotes and Prokaryotes cell. 4

Q- 2 Attempt any TWO of the following:

- A I Describe biosynthetic pathway of cholesterol. 6
- II Give classification of enzyme. 4
- B I Write a note on β -oxidation of fatty acids. 6
- II Explain: Induced fit theory of enzyme action 4
- C I Enlist factor affecting enzyme activity. Explain Michaelis-Menten constant
and its significations. 6
- II Define lipids and write functions of it in human body. 4

SECTION- II

Q- 3 Attempt any TWO of the following:

- | | | | |
|---|----|--|---|
| A | I | Define: (i) Transamination (ii) Carbohydrates (iii) Glycogenolysis (iv) Glycogenesis (v) Glycolysis (vi) Antiport system | 6 |
| | II | Classify protein based on chemical nature. | 4 |
| | | | |
| B | I | Explain TCA cycle. | 6 |
| | II | Give brief note on structure of cell membrane. | 4 |
| | | | |
| C | I | Discuss the process of active transport across cell membrane. | 6 |
| | II | Draw the pathway for cori cycle. | 4 |

Q- 4 Attempt any TWO of the following:

- | | | | |
|---|----|---|---|
| A | I | Discuss aerobic glycolysis with energetics. | 6 |
| | II | Explain oxidative deamination in brief. | 4 |
| | | | |
| B | I | Outline reactions of urea cycle. | 6 |
| | II | Describe α -helical structure of protein. | 4 |
| | | | |
| C | I | Draw a pathway for biogenesis of catecholamines. | 6 |
| | II | Write down short note on denaturation of protein. | 4 |